

12th and 13th April notes

Jenkins-Tomcat – EC2 instance

Class notes:

1. Install and setup Jenkins – on port 8080 – give SCM – Build it to get .war artifact
 2. Install Tomcat using wget and extract tar.gz -- also on 8080 → so change the port
- explore all the folders
 - webapps/... context.xml ---> keep one line
 - config/users.xml -- > remove all lines and add tomcat user with tomcat pwd
 - config/server.xml -- > esc line71 – change the port from 8080 to 8090 or any other
 - bin → ./startup.sh → Tomcat started

Manually == Chrome → Public ip:8090 – give .war file and deploy

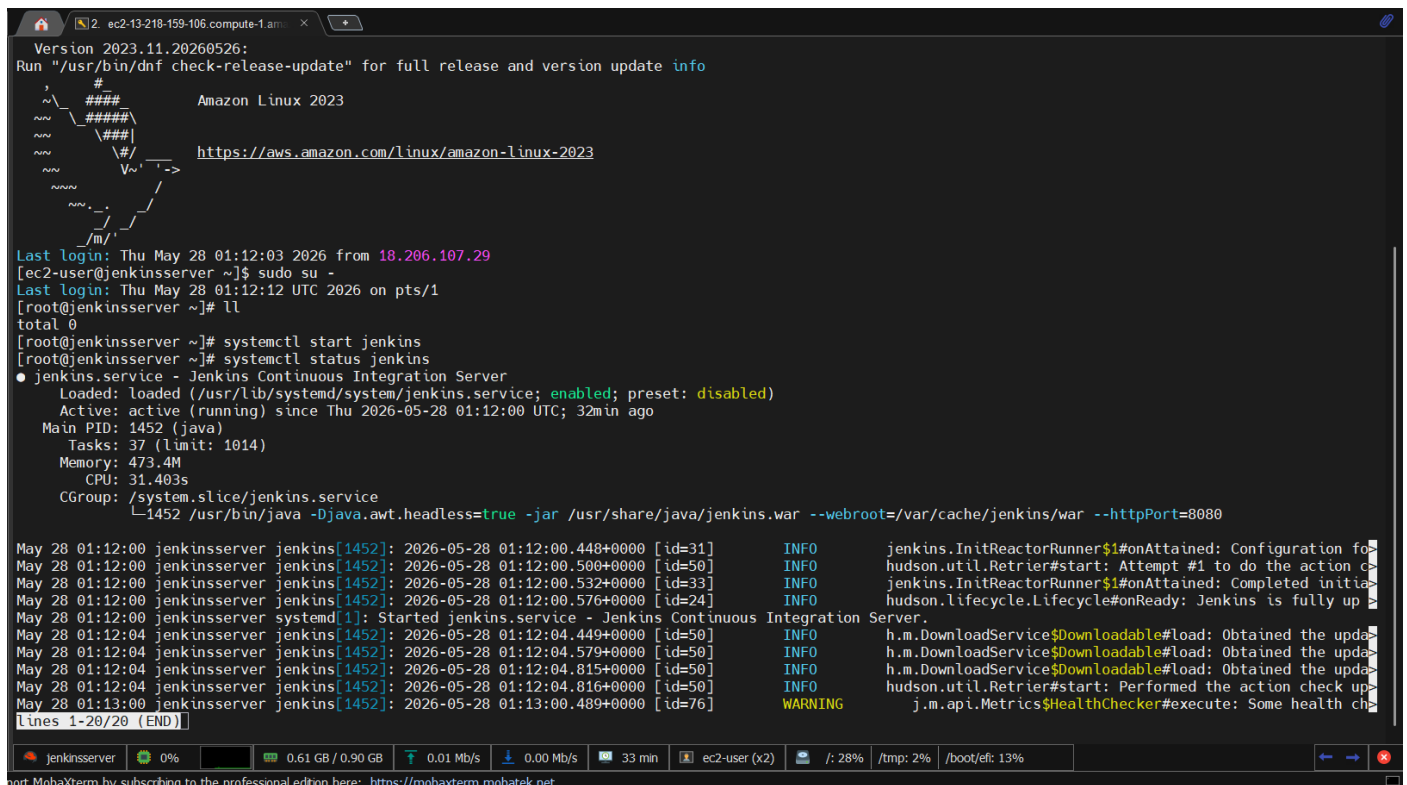
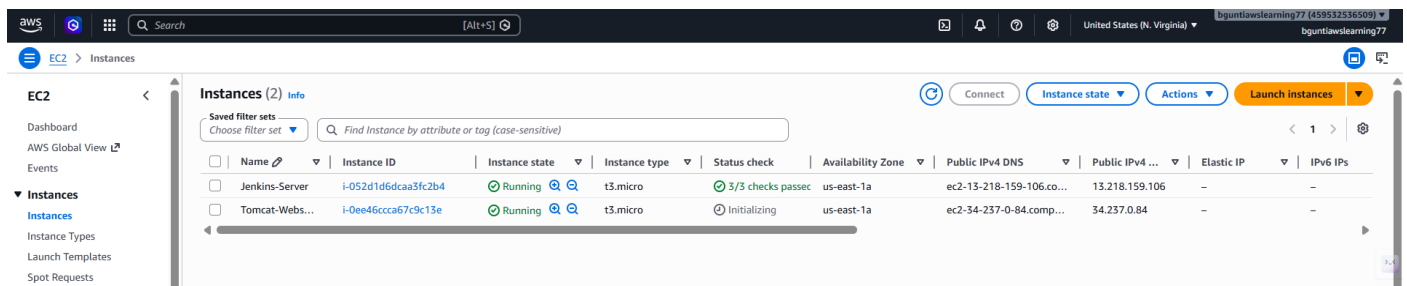
- logs/Catalina.out →

\$ cat catalina.out

\$ tail -100f Catalina.out

- webapps – ll → you can your deployed appln .war folder and its extraction folder → so that when public access your app, your app gets served from this extracted folder

Practical



← → ↻ Not secure 13.218.159.106:8080/job/B17dFacebook/ws/target/ ☆ 📁 🔍 Verify it's you ⋮

Jenkins / B17dFacebook ▾ / Workspace of B17dFace...

📄 Status **Workspace of B17dFacebook on Built-In Node**

🔗 Changes B17dFacebook / target / →

📁 Workspace 📁 facebookapp-4.0.0-SNAPSHOT

▶ Build Now 📁 maven-archiver

⚙️ Configure 📄 facebookapp-4.0.0-SNAPSHOT.war May 28, 2026, 1:52:18 AM 29.46 KiB 👁

🗑 Delete Project 📁 (all files in zip)

✎ Rename

Builds > ...

🔍 Filter /

Today

🟢 #1 1:52 AM ▾

Tomcat installed

Remote monitoring Follow terminal folder

tomcatserver 0% 0.27 GB / 0.90 GB 0.01 Mb/s 0.00 Mb/s 16 min ec2-user (x2) /: 26% /tmp: 0% /boot/efi: 13%

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Thunderstorm w...
In effect

27°C Sunny

Remote monitoring Follow terminal folder

```
-rw-r-----. 1 root root 1333 May  5 13:33 context.xml
[root@tomcatserver META-INF]# pwd
/opt/apache-tomcat-10.1.55/webapps/manager/META-INF
[root@tomcatserver META-INF]# ll
total 4
-rw-r-----. 1 root root 1333 May  5 13:33 context.xml
[root@tomcatserver META-INF]#
```

tomcatserver 0% 0.28 GB / 0.90 GB 0.01 Mb/s 0.00 Mb/s 18 min ec2-user (x2) /: 26% /tmp: 0% /boot/efi: 13%

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34.237.0.84 (ec2-user)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

/home/ec2-user/

Name

..

.ssh

.bashrc

.bash_profile

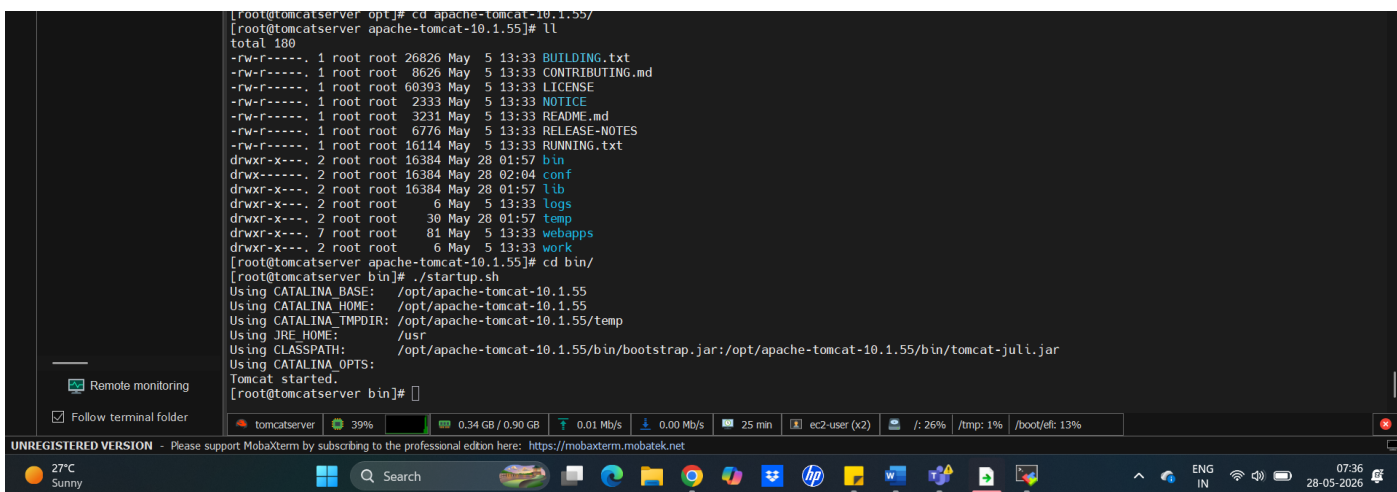
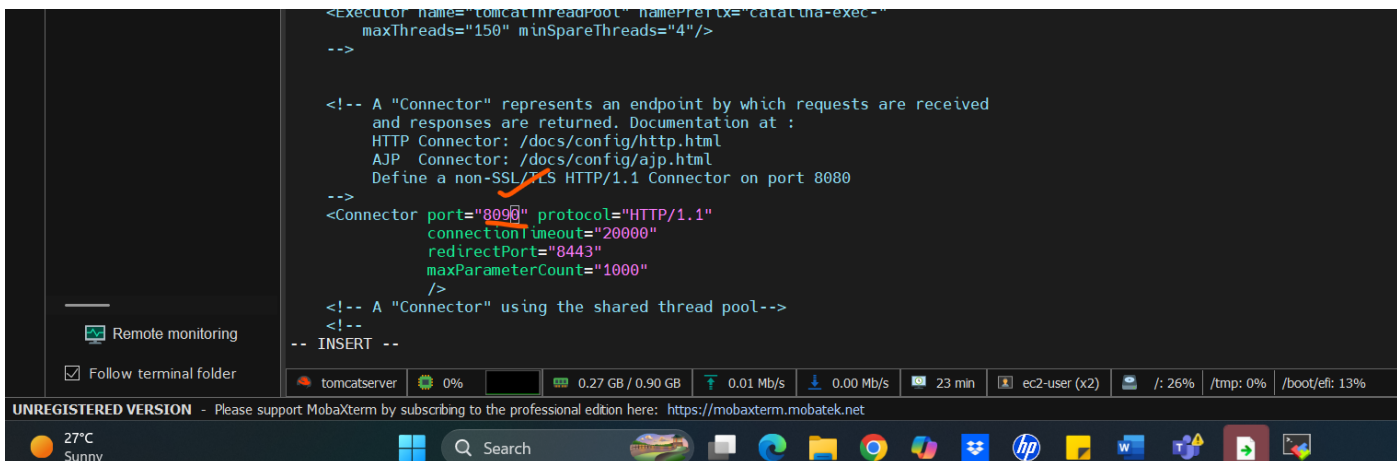
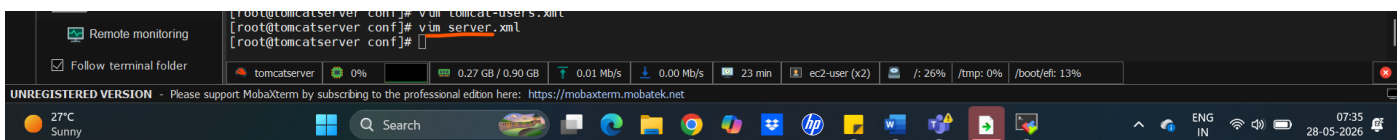
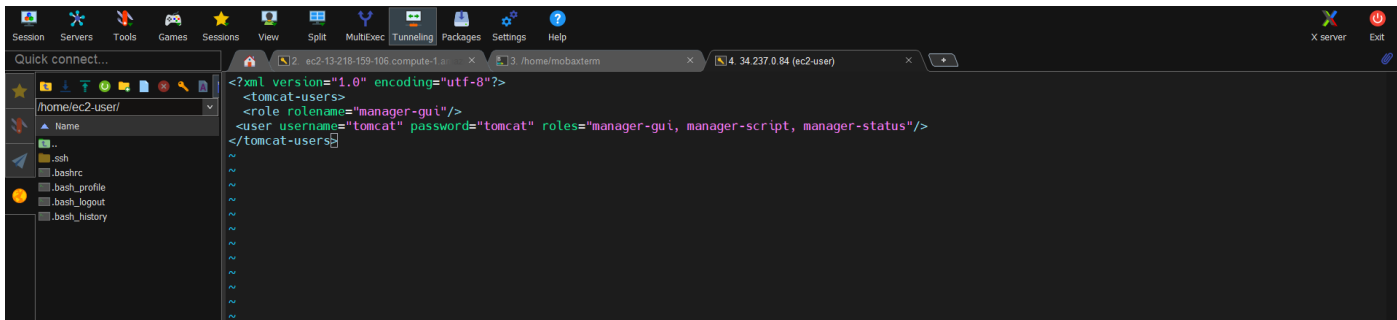
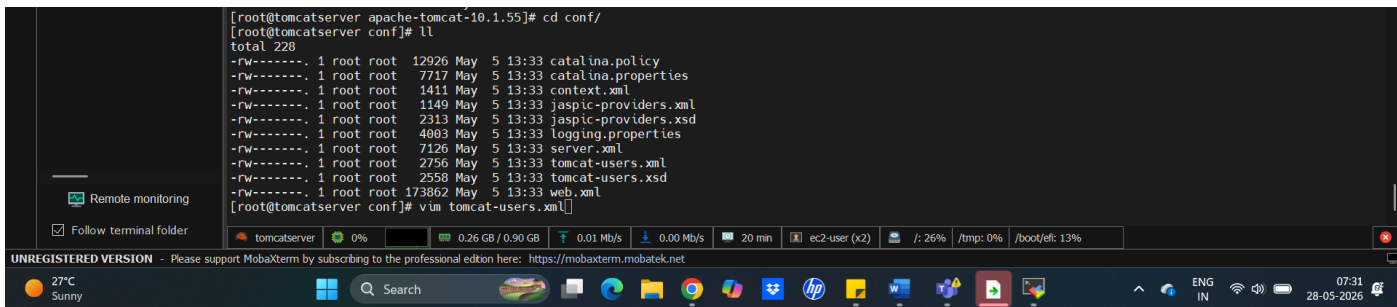
.bash_logout

.bash_history

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements.  See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License.  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  </Context>
~
~
~
```



```
[root@tomcatserver opt]# cd apache-tomcat-10.1.55/
[root@tomcatserver apache-tomcat-10.1.55]# ll
total 180
-rw-r--r-- 1 root root 26826 May 5 13:33 BUILDING.txt
-rw-r--r-- 1 root root 8626 May 5 13:33 CONTRIBUTING.md
-rw-r--r-- 1 root root 60393 May 5 13:33 LICENSE
-rw-r--r-- 1 root root 2333 May 5 13:33 NOTICE
-rw-r--r-- 1 root root 3231 May 5 13:33 README.md
-rw-r--r-- 1 root root 6776 May 5 13:33 RELEASE-NOTES
-rw-r--r-- 1 root root 16114 May 5 13:33 RUNNING.txt
drwxr-x--- 2 root root 16384 May 28 01:57 bin
drwxr-x--- 2 root root 16384 May 28 02:04 conf
drwxr-x--- 2 root root 16384 May 28 01:57 lib
drwxr-x--- 2 root root 6 May 5 13:33 logs
drwxr-x--- 2 root root 30 May 28 01:57 temp
drwxr-x--- 7 root root 81 May 5 13:33 webapps
drwxr-x--- 2 root root 6 May 5 13:33 work
[root@tomcatserver apache-tomcat-10.1.55]# cd bin/
[root@tomcatserver bin]# ./startup.sh
Using CATALINA_BASE: /opt/apache-tomcat-10.1.55
Using CATALINA_HOME: /opt/apache-tomcat-10.1.55
Using CATALINA_TMPDIR: /opt/apache-tomcat-10.1.55/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/apache-tomcat-10.1.55/bin/bootstrap.jar:/opt/apache-tomcat-10.1.55/bin/tomcat-juli.jar
Tomcat started.
[root@tomcatserver bin]#
```

← → ↻ ⚠ Not secure 34.237.0.84:8090/manager/html

/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy
					Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy
					Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy
					Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy
					Expire sessions with idle ≥ 30 minutes

Deploy
Deploy directory or WAR file located on server

Context Path:

Version (for parallel deployment):

XML Configuration file path:

WAR or Directory path:

Deploy

WAR file to deploy

Select WAR file to upload No file chosen

Deploy

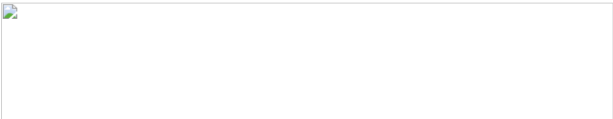
Configuration
Re-read TLS configuration files

TLS host name (optional)

Re-read

← → ↻ ⚠ Not secure 34.237.0.84:8090/facebookapp-4.0.0-SNAPSHOT%20(1)/

MY NAME IS MADHUKIRAN GOREKAR , A PASSIONATE DEVOPS TRAINER & CONTINUOUS LEARNER



- Now, let us see how to automate using 'Jenkins'
- 1. you can see mindcircuit17d code folder in github
- 2. Jenkins – new Item – facebook-prod-job-freestyle project-ok
- SCM – give https link of code – main brabch
- \$ Yum install git -y in your ec2
- Env -delete workspace, add timestap
- Build steps – Execute shell –mvn clean install
- Build now
- console output
- \$ yum install maven -y
- Build now => workspace => target/.war

Manually – upload .war file in Tomcat

Automatically :

- Post-Build action – cant see Deploy function → Extra functionality – need plugin

- Manage – Avail – Deploy to container – install

- Post-Build action – Add – Deploy war to a container

War/ear files : */.war

Context path :b17-facebook

Container – Add – Tomcat 9.x

Tomcat Url – http://:8080

Cred – add --- User with passwd – next – username, pwd, id, desc ==tomcat

Enable – deploy on failure

Save

- Build now

Tomcat – refresh – you can your appln deployed

\$ webapps→ ll -> b17facebook

\$ cd logs →tail -1000f Catalina.out

- If code changed in github

- Build now

- Tomcat – refresh – you can your appln deployed with changes

The screenshot shows the Jenkins 'Configure' page for a job named 'B17dFacebook'. The left sidebar has a menu with 'General', 'Source Code Management', 'Triggers', 'Environment' (selected), 'Build Steps', and 'Post-build Actions'. The main content area is titled 'Configure' and has a sub-section 'Environment'. Below this, there are several checkboxes: 'Delete workspace before build starts' (checked), 'Use secret text(s) or file(s)' (unchecked), 'Add timestamps to the Console Output' (checked), 'Inspect build log for published build scans' (unchecked), 'Terminate a build if it's stuck' (unchecked), and 'With Ant' (unchecked). Below the 'Environment' section is the 'Build Steps' section, which contains a single step named 'Execute shell'. The command for this step is 'mvn clean install'.

The screenshot shows the Jenkins 'Manage Jenkins' page, specifically the 'Plugins' section. The left sidebar has a menu with 'Updates' (11), 'Available plugins' (selected), 'Installed plugins', and 'Advanced settings'. The main content area is titled 'Plugins' and has a search bar with the text 'Deploy'. Below the search bar is a table of available plugins. The table has columns for 'Install', 'Name', 'Released', and 'Health'. The first plugin is 'Deploy to container' (version 1.17), which is already installed (checked box) and has a health status of 89. The second plugin is 'Docker Pipeline' (version 634.vedc7242b_eda_7), which is not installed (unchecked box) and has a health status of 76. There is a note at the bottom of the 'Docker Pipeline' row: 'This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.'

Install	Name	Released	Health
☒	Deploy to container 1.17 Artifact Uploaders This plugin allows you to deploy a war to a container after a successful build. Glassfish 3.x remote deployment	1 yr 0 mo ago	89
☐	Docker Pipeline 634.vedc7242b_eda_7 pipeline DevOps Deployment docker Build and use Docker containers from pipelines. This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	7 mo 16 days ago	76

 **Jenkins** / B17dFacebook ▾ / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

EXECUTE SHELL COMMAND

Command

[See the list of available environment variables](#)

Filter

Aggregate downstream test results

Archive the artifacts

Build other projects

Publish JUnit test result report

Record fingerprints of files to track usage

Git Publisher

Deploy war/ear to a container ✓

E-mail Notification

Editable Email Notification

Set GitHub commit status (universal)

Set build status on GitHub commit [deprecated]

Delete workspace when build is done

+ Add post-build action

... sending notifications, archiving artifacts, or triggering other jobs.

Save

Apply

 **Jenkins** / B17dFacebook ▾ / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions**

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Deploy war/ear to a container

WAR/EAR files ?

**/*.war

War/ear files to deploy. Relative to the workspace root. You can also specify Ant-style GLOBs, like "**/*.war" (from Deploy to container Plugin)

Context path ?

b17dFacebook

Containers

Tomcat 9.x Remote

Credentials

tomcat/***** (tomcat) ▾ + Add

Tomcat URL ?

http://34.237.0.84:8090/

Advanced ▾

+ Add Container

Save

Apply

 **Jenkins**

[/ B17dFacebook](#) [Workspace of B17dFace...](#)

Status

Changes

Workspace

Wipe Out Current Workspace

Build Now

Configure

Delete Project

Rename

Workspace of B17dFacebook on Built-In Node

B17dFacebook /

 .git

 src/main/webapp

 target

 Dockerfile

 Jenkinsfile

 Jenkinsfile_docker_session

 Jenkinsfile_env_vars

 pom.xml

 README.md

May 28, 2026, 2:16:23 AM

May 28, 2026, 2:16:23 AM

May 28, 2026, 2:16:23 AM

May 28, 2026, 2:16:23 AM

May 28, 2026, 2:16:23 AM

May 28, 2026, 2:16:23 AM

May 28, 2026, 2:16:23 AM

May 28, 2026, 2:16:23 AM

272 B

690 B

738 B

1.10 KiB

1.27 KiB

16 B













[\(all files in zip\)](#)

Builds >

Filter

Today

 #2 2:16 AM

 #1 1:52 AM

[←](#) [→](#) [↺](#) [⚠ Not secure](#) 13.218.159.106:8080/job/B17dFacebook/2/console

 **Jenkins**

[/ B17dFacebook](#) [#2](#)

Timestamps

View as plain text

☒ System clock time

☐ Use browser timezone

☐ Blipped time

☐ None

02:16:26 [WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!

02:16:26 [INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/B17dFacebook/src/test/resources

02:16:26 [INFO]

02:16:26 [INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ facebookapp ---

02:16:26 [INFO] No sources to compile

02:16:26 [INFO]

02:16:26 [INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ facebookapp ---

02:16:26 [INFO] No tests to run.

02:16:26 [INFO]

02:16:26 [INFO] --- maven-war-plugin:3.3.2:war (default-war) @ facebookapp ---

02:16:27 [INFO] Packaging webapp

02:16:27 [INFO] Assembling webapp [facebookapp] in [/var/lib/jenkins/workspace/B17dFacebook/target/facebookapp-4.0.0-SNAPSHOT]

02:16:27 [INFO] Processing war project

02:16:27 [INFO] Copying webapp resources [/var/lib/jenkins/workspace/B17dFacebook/src/main/webapp]

02:16:27 [INFO] Building war: /var/lib/jenkins/workspace/B17dFacebook/target/facebookapp-4.0.0-SNAPSHOT.war

02:16:27 [INFO]

02:16:27 [INFO] --- maven-install-plugin:2.4:install (default-install) @ facebookapp ---

02:16:27 [INFO] Installing /var/lib/jenkins/workspace/B17dFacebook/target/facebookapp-4.0.0-SNAPSHOT.war to /var/lib/jenkins/.m2/repository/com/example/facebookapp/4.0.0-SNAPSHOT/facebookapp-4.0.0-SNAPSHOT.war

02:16:27 [INFO] -----

02:16:27 [INFO] BUILD SUCCESS

02:16:27 [INFO] -----

02:16:27 [INFO] Total time: 2.400 s

02:16:27 [INFO] Finished at: 2026-05-28T02:16:27Z

02:16:27 [INFO] -----

02:16:27 [INFO] [DeployPublisher][INFO] Attempting to deploy 1 war file(s)

02:16:27 [DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/B17dFacebook/target/facebookapp-4.0.0-SNAPSHOT.war to container Tomcat 9.x Remote with context b17dFacebook

02:16:27 [INFO] [/var/lib/jenkins/workspace/B17dFacebook/target/facebookapp-4.0.0-SNAPSHOT.war] is not deployed. Doing a fresh deployment.

02:16:27 Deploying [/var/lib/jenkins/workspace/B17dFacebook/target/facebookapp-4.0.0-SNAPSHOT.war]

02:16:27 Finished: SUCCESS

Tomcat Web Application Manager

Message: OK

Manager

List Applications

HTML Manager Help

Manager Help

Server Status

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/facebookapp-4.0.0-SNAPSHOT(1)	None specified	Archetype Created Web Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Deploy

Deploy directory or WAR file located on server

Mindcircuit 16d and mindcourcuit 17d code applications

← → ↻ Not secure 34.237.0.84:8090/manager/html ☆ 📄 📄 📄 📄 Verify it's you

Manager

List Applications HTML Manager Help Manager Help Server Status

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/b17dFacebook	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/facebookapp-4.0.0-SNAPSHOT (1)	None specified	Archetype Created Web Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Deploy

Deploy directory or WAR file located on server

Context Path:

Version (for parallel deployment):

```
[root@tomcatserver bin]# cd ..
[root@tomcatserver apache-tomcat-10.1.55]# ll
total 196
-rw-r-----. 1 root root 26826 May  5 13:33 BUILDING.txt
-rw-r-----. 1 root root  8626 May  5 13:33 CONTRIBUTING.md
-rw-r-----. 1 root root 60393 May  5 13:33 LICENSE
-rw-r-----. 1 root root  2333 May  5 13:33 NOTICE
-rw-r-----. 1 root root  3231 May  5 13:33 README.md
-rw-r-----. 1 root root  6776 May  5 13:33 RELEASE-NOTES
-rw-r-----. 1 root root 16114 May  5 13:33 RUNNING.txt
drwxr-x---. 2 root root 16384 May 28 01:57 bin
drwx-----. 3 root root 16384 May 28 02:06 conf
drwxr-x---. 2 root root 16384 May 28 01:57 lib
drwxr-x---. 2 root root  162 May 28 02:08 logs
drwxr-x---. 2 root root   30 May 28 01:57 temp
drwxr-x---. 9 root root 16384 May 28 02:16 webapps
drwxr-x---. 3 root root   22 May 28 02:06 work
[root@tomcatserver apache-tomcat-10.1.55]# cd logs/
[root@tomcatserver logs]# ll
total 28
-rw-r-----. 1 root root 7673 May 28 02:16 catalina.2026-05-28.log
-rw-r-----. 1 root root 8084 May 28 02:16 catalina.out
-rw-r-----. 1 root root  408 May 28 02:06 localhost.2026-05-28.log
-rw-r-----. 1 root root 2134 May 28 02:18 localhost_access_log.2026-05-28.txt
-rw-r-----. 1 root root 1668 May 28 02:16 manager.2026-05-28.log
[root@tomcatserver logs]# tail -1000f catalina.out
```

Remote monitoring

Follow terminal folder

tomcatserver 0% 0.40 GB / 0.90 GB 0.01 Mb/s 0.00 Mb/s 39 min ec2-user (x2) /: 26% /tmp: 1% /boot/efi: 13%

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28°C Mostly sunny

```
[root@tomcatserver apache-tomcat-10.1.55]# ll
total 196
-rw-r-----. 1 root root 26826 May  5 13:33 BUILDING.txt
-rw-r-----. 1 root root  8626 May  5 13:33 CONTRIBUTING.md
-rw-r-----. 1 root root 60393 May  5 13:33 LICENSE
-rw-r-----. 1 root root  2333 May  5 13:33 NOTICE
-rw-r-----. 1 root root  3231 May  5 13:33 README.md
-rw-r-----. 1 root root  6776 May  5 13:33 RELEASE-NOTES
-rw-r-----. 1 root root 16114 May  5 13:33 RUNNING.txt
drwxr-x---. 2 root root 16384 May 28 01:57 bin
drwx-----. 3 root root 16384 May 28 02:06 conf
drwxr-x---. 2 root root 16384 May 28 01:57 lib
drwxr-x---. 2 root root  162 May 28 02:08 logs
drwxr-x---. 2 root root   30 May 28 01:57 temp
drwxr-x---. 9 root root 16384 May 28 02:16 webapps
drwxr-x---. 3 root root   22 May 28 02:06 work
[root@tomcatserver apache-tomcat-10.1.55]# cd webapps/
[root@tomcatserver webapps]# ll
total 96
drwxr-x---. 3 root root 16384 May 28 01:57 ROOT
drwxr-x---. 4 root root   54 May 28 02:16 b17dFacebook
-rw-r-----. 1 root root 30163 May 28 02:16 b17dFacebook.war
drwxr-x---. 16 root root 16384 May 28 01:57 docs
drwxr-x---. 7 root root   99 May 28 01:57 examples
drwxr-x---. 4 root root   54 May 28 02:09 'facebookapp-4.0.0-SNAPSHOT (1)'
-rw-r-----. 1 root root 30163 May 28 02:09 'facebookapp-4.0.0-SNAPSHOT (1).war'
drwxr-x---. 6 root root   79 May 28 01:57 host-manager
drwxr-x---. 6 root root  114 May 28 01:57 manager
[root@tomcatserver webapps]#
```

Remote monitoring

Follow terminal folder

tomcatserver 0% 0.39 GB / 0.90 GB 0.01 Mb/s 0.00 Mb/s 42 min ec2-user (x2) /: 26% /tmp: 1% /boot/efi: 13%

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